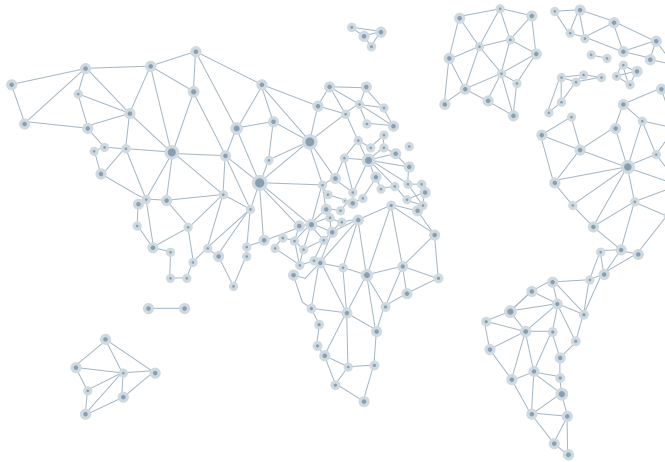
CS 260 - Data Structures

Syllabus

Yusuf Osmanlioglu
Department of Computer Science
Drexel University



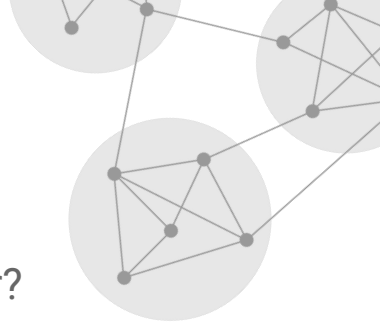
Team

- Instructor: Yusuf Osmanlioglu, PhD.,
  - yo42@drexel.edu, Office: 1145
- TA: Juwhan “Isaac” Jung
  - jj984@drexel.edu, office hours at CLC
- Office hours

Monday	Tuesday	Wednesday	Thursday	Friday
	Isaac: 11 AM-1 PM (CLC)	**Yusuf:** 3-4 PM (Office 1145)	**Yusuf:** 3-4 PM (Office 1145)	

- Make sure to reach out to us to get help…
  - … sooner rather than later!!!
- You can also post questions to Discord.
  - You will get answers within 24 hours in weekdays
  - Do not expect response over the weekends.

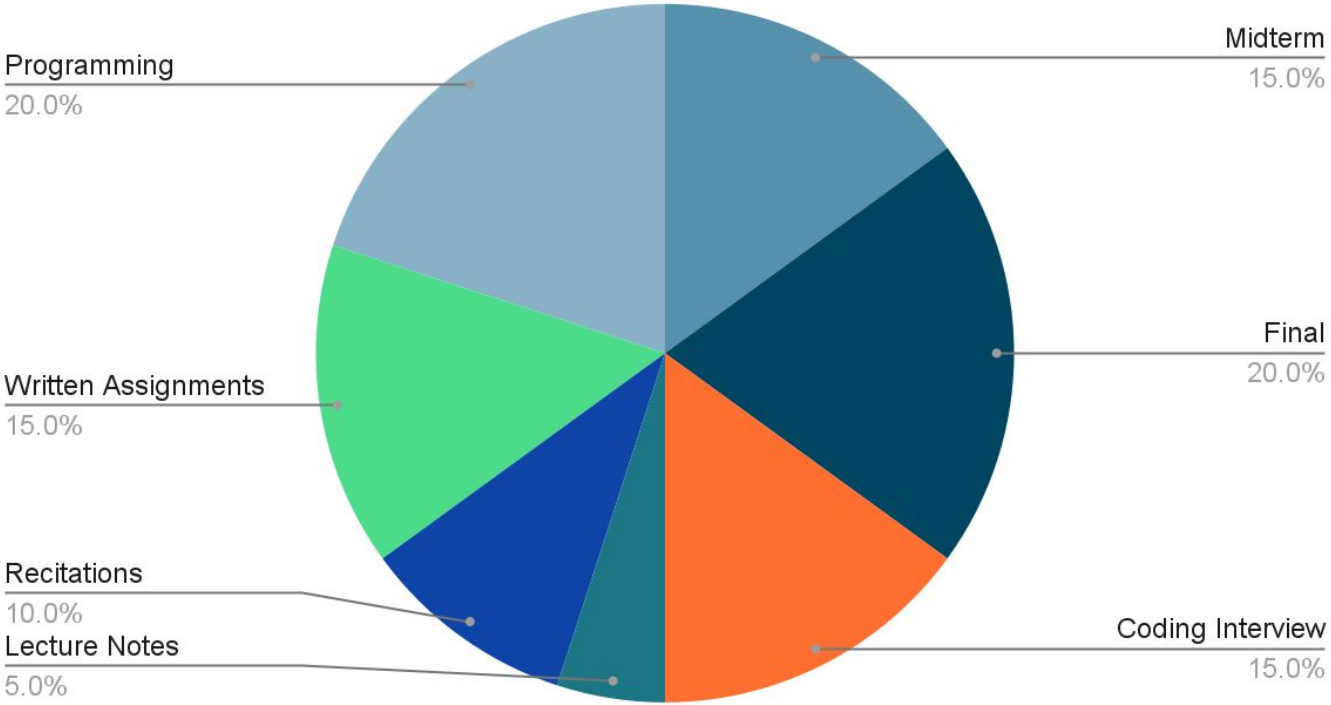
Goals



- Given a tough real life problem, how would you solve it using a computer?
  - How do you make an abstraction of the problem?
  - How do you represent the problem data in computer?
  - How do you solve the problem?
- The goal is to make you become “problem solvers”.
- This class will make you gain skills that will be useful for:
  - Your professional career in the long run
  - Also in your coding interviews, which are right across the corner!

Class Setup

Points scored

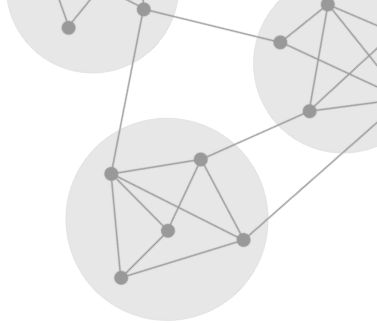


Reading Assignments
3% extra credit

Main Class Setup

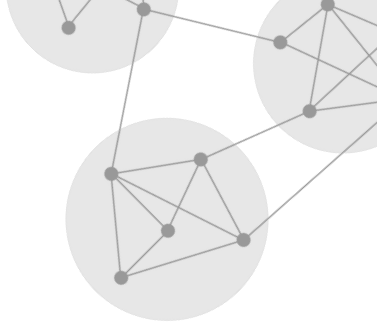
- Readings (extra credit)
  - There will be assigned readings for most of the weeks, along with a small set of questions.
  - Submit your handwritten answers to gradescope (via Blackboard).
  - Grading will be based on effort.
- Lecture Notes
  - You will take notes during the lecture (**hand written** on a sheet of paper!), and submit the picture of your notes to gradescope (via blackboard) at the end of the lecture.
  - TAs will grade out of 5 points based on in person attendance (2 pt) and quality of notes (3pt)
  - **Lowest two** grades will be dropped
- Recitations
  - We will solve problems in open ended quiz format. You will do it in groups.
  - TAs will grade based on effort, **lowest** grade will be dropped
- Exams
  - Midterm at week 6, final at week 12 (which will be cumulative)

Main Class Setup



- Coding Interview
  - A single 1 hour in person mock-up coding interview
  - Will be scheduled throughout weeks 7-10
  - To measure your problem solving skills and justifications for how you solve problems
- Written assignments
  - You'll have four of these
  - Will be very similar to exam questions
  - Late submission allowed up to two days with 10 points off for each late day. No submission beyond two late days.
- Programming Assignments
  - There will be 5 programming assignments in total, with occasional extra credit component!
  - You will be given a real life problem to solve
    - It's solution will involve the data structures and algorithms that we learn in the class
  - First programming assignment is already posted!!

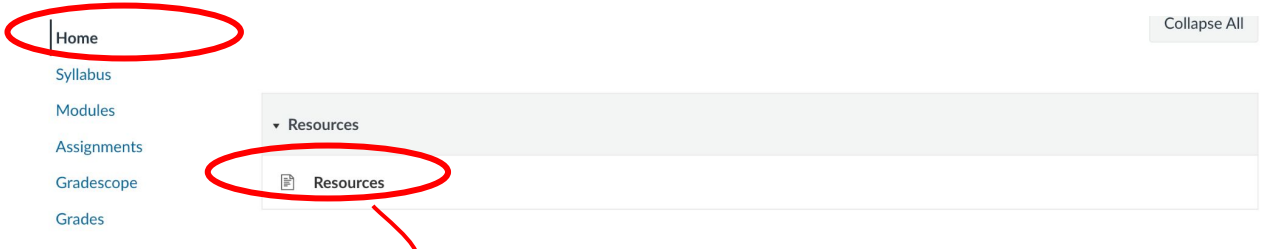
Programming Assignments



- Assignments will be done in **C**!
  - WHY?
    - High level languages (like Python) hides many necessary details!!
    - Performance is an important part of programming, where the data structure and algorithm you use aims to improve. C is a perfect environment to understand what improves performance.
- Implementations are **to be done on tux**!
  - You can use Visual Studio on your local machine to connect to tux!
- **Late Submission**:
  - You are allowed to submit programming assignments **up to 2 days late**, with 10/100 points **late penalty** for each day.
  - No later submissions will be accepted beyond 2 days!

Programming Assignments

- There are 7 videos to hone your C skills! Make sure to go through them!
  - Links are provided in the Canvas, Resources module, C Programming link.



Resources

Make sure to check announcements and the syllabus. Course content and the assignments are listed under modules link to the left.

Below are general resources that you might find useful:

- C Programming : C programming support (necessary for programming assignments and mock coding interviews)
- Latex : How to use Latex (necessary for written assignments)
- Suggestions from Past Students : No need to reinvent the wheel!
- Schedule : 10 week terms at Drexel is too quick! Plan ahead of time...
- Office Hours : Attending office hours will make the class way easier
- syllabus.pdf

C Programming

Prof. Galen Long's CS265 C Textbook

Link here: https://www.cs.drexel.edu/~nkl43/cs265/c/

Recorded Videos on Key C Concepts

Here is a playlist specifically prepared for this course. These videos will help you quickly prepare for the programming assignments. They were created by Jonathan Parlett, who at the top of his class when he took CS 260. The playlist includes videos on:

- Memory in C
- Pointers and Memory
- Dynamic Array Programming Example
- Advanced Topics in C
- Dictionary Programming Example
- Valgrind and Makefiles
- VSCode Remote Development Tux Setup

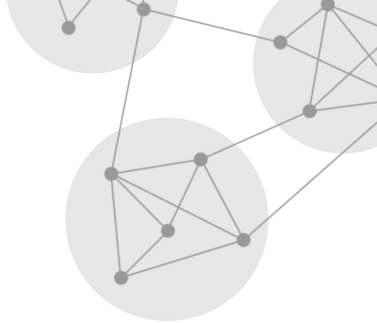
This is probably the quickest and most efficient way for you to learn about the necessary C skills for this course.

Dynamic Memory Management

You will encounter many segmentation faults when coding in C. Use Valgrind to detect where you are leaking memory.

Additionally, here is a useful tool to visualize the memory usage of your program, which might help in getting a better understanding of pointers.

Logistics



- Gradescope with Canvas integration
  - For submitting assignments, we will use Canvas/gradescope mixture.
- Discord
  - For questions/comments

Cheating

- A BIG NO NO!!!
- Cheating includes: taking answers from
  - another person,
  - a website,
  - ChatGPT,
  - Copilot etc.
- **Discussing an idea** on the whiteboard **is not considered cheating** and is encouraged.
- Academic Honesty Violations **will be reported to the University**.
  - No exception!
- Punishment will be determined by the severity of the incident.
- Punishments **at least include**, but are not limited to,
  - **-30/100 on Assignment/Exam Violation took place on**
  - Failing grade for class
  - Deduction of one letter grade



Use of AI

- Artificial intelligence tools such as large language models **are not permitted to be used in preparing submitted work** for this course.
- This includes but are not limited to the use of:
  - large language models (such as ChatGPT) for written assignments
  - code aiding tools such as Copilot.
- Students are expected to comfortably be able to explain what they have submitted as their own work.



ChatGPT

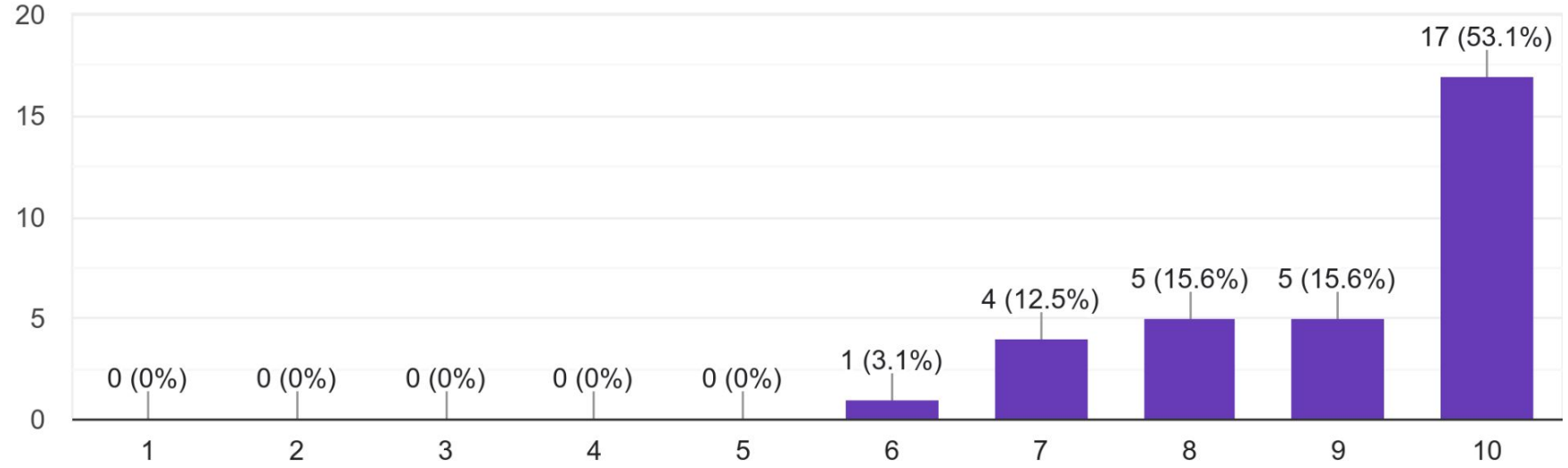


GitHub Copilot

Review notes from prior students

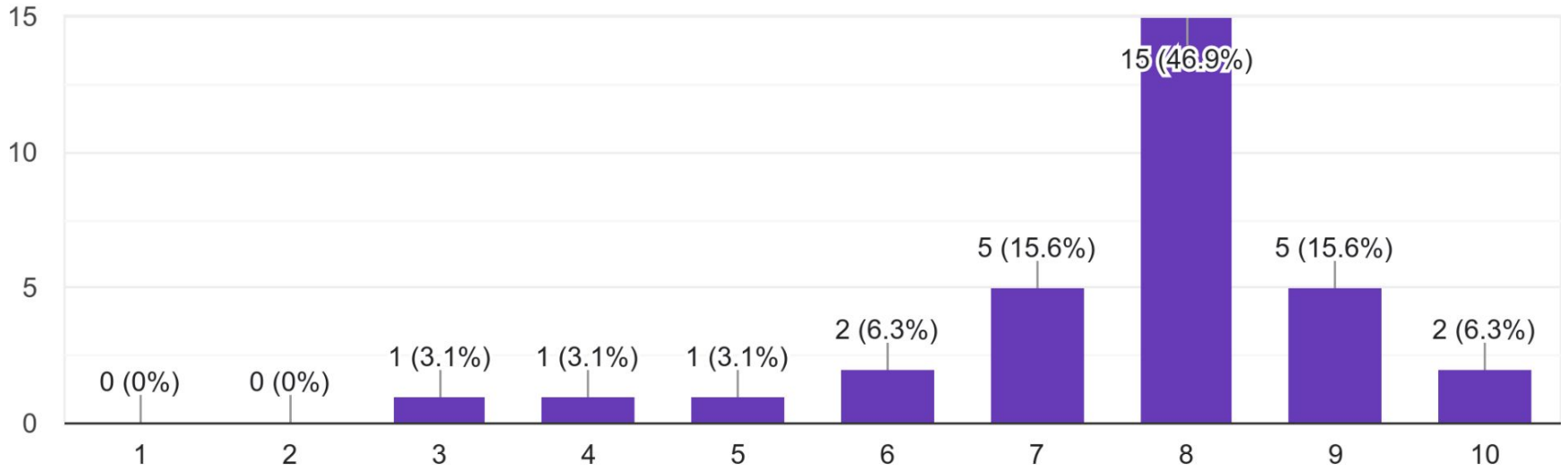
How important do you think is this class for your career?

32 responses

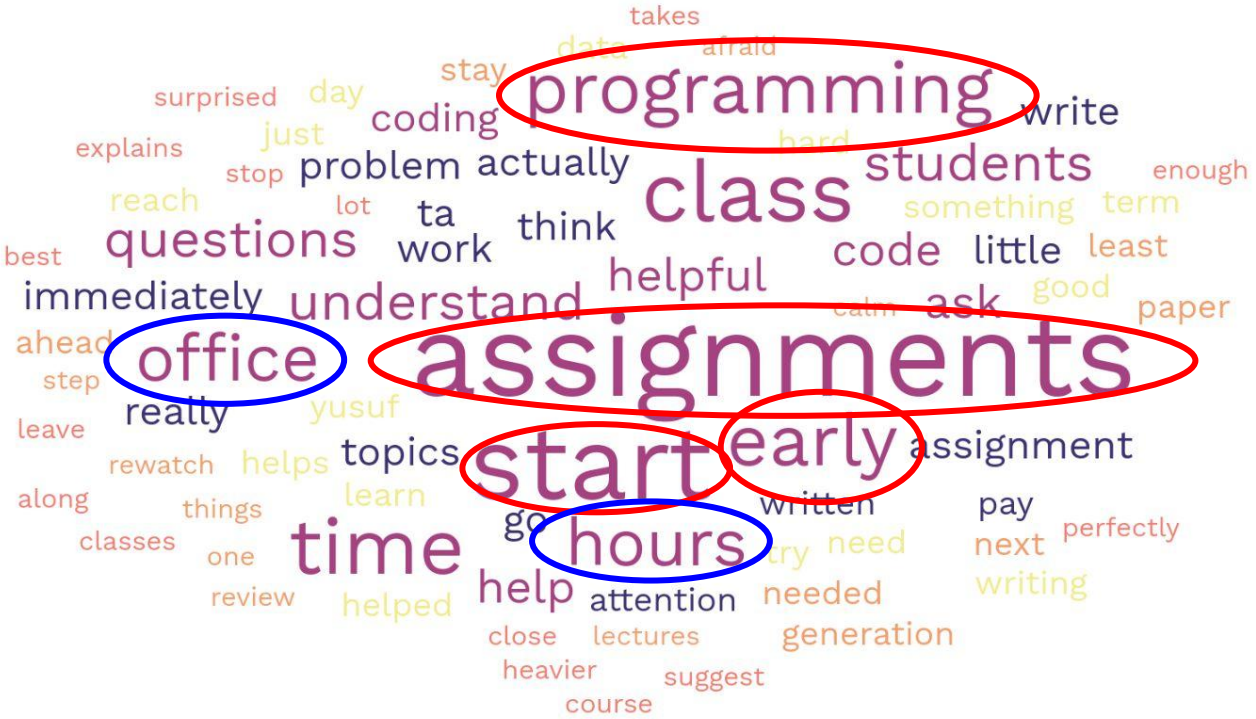


How hard is this class?

32 responses



Suggestions to the next generation of students taking class in this format



Suggestions to the next generation of students taking class in this format

- Start early to programming assignments:
  - I would suggest to start the assignments early and do them in smaller chunks. There is no way to complete an assignment all at once.
  - My main suggestion would be to start the programming assignments earlier than you think you need to. Something will always go wrong and it will take extra time to solve it.
  - Start early on the programming assignments and make sure to make each minute of office hours worth everyone's time.
  - Start the assignments early. I would start them early each time and be done with plenty of time.
  - Start PA's early.
  - ….
- Watch prep C videos before classes
- Commit time to this course if you can, don’t take too many difficult courses with this course
  - Either prioritize this class, or ensure you aren't taking other classes that also require a heavy amount of course work.
- Make the most use of office hours
  - Plan working on assignments around office hours and start early so that if you get stuck you can ask the instructor or TA for help.
  - Reach out to professor or TA when stuck.